Aurora Express Wheel Assembly

Bottleneck Analysis & Efficiency Improvement Recommendation

Identified Bottleneck

Element 14 – **"Mount wheel to axle"** has the highest average cycle time of all 20 elements in the assembly process, at **5 minutes 24 seconds** per unit - 40 seconds longer than the next-slowest step ("Conduct wheel alignment," 4:44). Time-study notes flag this task as the heaviest component in the process and dependent on crane availability, meaning its duration is tied to shared equipment rather than operator skill alone. As the slowest and most resource-constrained step, it caps the throughput of the entire line and creates a queue for every downstream task (locking mechanism, alignment, functional test, transport).

Actionable Suggestion

Dedicate a lifting fixture and pre-stage the axle for this step, instead of relying on a shared/general-purpose crane.

- Assign a dedicated hoist or a fixed overhead lifting point at the mounting station so operators never wait for a shared crane to become free.
- Pre-position and align the axle on a jig immediately before the mounting step, so the crane is used only for the final lift-and-fit, not for search, positioning, or repositioning.
- Introduce a simple visual/mechanical alignment guide (e.g., locating pins or a guided rail) to reduce the fine positioning time during the lift.

Expected Impact on the Bottleneck and Overall Process

Removing crane contention directly attacks the two drivers of this step's length: equipment wait time and handling precision. A dedicated fixture eliminates the variable wait for a shared crane, which is the main source of the spread seen in the observed times (5:12 to 5:35 across trials — a 23-second range). Pre-staging the axle shifts preparation work outside the timed lift, so the crane is engaged only for the core mounting action.

Even a conservative 60–90 second reduction per unit on this step would lower the cycle time of the single longest task in the process, which in turn raises the line's maximum throughput, since cycle time for the whole assembly is set by its slowest step. Because every subsequent operation (steps 15–20: locking, alignment, testing, cleanup, documentation, transport) queues behind this one, a faster and more consistent mount time also reduces idle time for downstream operators and shortens total lead time per wheel assembly.

In short: this change targets the largest single time sink in the process and its main source of variability, giving the clearest available lever for improving overall assembly throughput.